Dennis Menn

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RESEARCH INTERESTS

Generative video and interactive world models, focusing on efficiency and real-time generation.

EDUCATION

Ph.D. in Electrical and Computer Engineering, University of Texas at Austin — Aug 2023 - Expected Aug 2027
Advisor: Prof. Diana Marculescu

M.S. in Electrical and Computer Engineering, National Taiwan University — Jul 2023
Advisor: Prof. Lin-shan Lee

B.S. in Electrical and Computer Engineering, National Taiwan University — Jun 2021

EXPERIENCE

Morpheus @ Roblox - Research Intern — May 2026 - Dec 2026
Mentors: Tianpei Gu and Xun Huang

- Built **MPEG-Diffusion**, which generates the **visually important region at high resolution** and the rest at low resolution: **up to 4.8× faster** while maintaining **high visual quality**, with a **self-selected** high-resolution region.

Futurewei - Research Intern (Austin) — May 2025 - Aug 2025

- Proposed a **content-aware video editing pipeline** that reuses edited duplicated tokens.

Energy-Aware Computing Lab, UT Austin - Graduate Research Assistant — Jan 2024 - Present

- Developed **LIPAR**, a training-free framework that adapts pixel-space compression techniques to latent pruning for video generation, achieving an average **1.53× throughput boost** (19.3 FPS on an RTX 4090) and a 20% memory reduction without compromising visual quality.
- Built **Streaming LIPAR**: real-time video editing at **20 FPS** on a single RTX 4090 (4-step denoising).

PUBLICATIONS

- "MPEG-Diffusion: Mixed-Resolution Video Generation." **Dennis Menn**, Tianpei Gu, Guande He, Aryan V S, Yanke Song, Nan Liu, Desai Xie, Xun Huang, Chenfeng Xu, Diana Marculescu. **Under submission, CVPR 2027**. arXiv:2611.12069.
- "Latent Inter-Frame Pruning with Attention Recovery." **Dennis Menn**, Yuedong Yang, Bokun Wang, Xiwen Wei, Mustafa Munir, Feng Liang, Radu Marculescu, Chenfeng Xu, Diana Marculescu. **NeurIPS 2026**.
- "Similarity Trajectories: Linking Sampling Process to Artifacts in Diffusion-Generated Images." **Dennis Menn**, Feng Liang, Hung-Yueh Chiang, Diana Marculescu. **WACV Workshops 2025**.
- "Synthesizing Artifact Dataset for Pixel-level Detection." **Dennis Menn**, Feng Liang, Diana Marculescu. Preprint, arXiv:2509.19589.
- "Investigating Human-Identifiable Features Hidden in Adversarial Perturbations." **Dennis Menn**, Tzu-hsun Feng, Sriram Vishwanath, Hung-yi Lee. Preprint, arXiv:2309.16878.

INVITED TALKS

Streaming LIPAR: Training-Free Real-Time Video Editing - Daydream Scope Workshop — Apr 2026

SKILLS

Languages: Python, C/C++, MATLAB, Shell

Tools: PyTorch, scikit-learn, OpenCV, SQL, HLS, Quartus, PSpice